DO NOW — RECALL

Lesson 136 · Normalisation: First and Second Normal Fo

Name: _____ Class: _____ Date: _____

1. What table resolves a many-to-many database relationship?

2. What is a composite key?

3. When does a linear search stop?

4. In Big O notation, what is binary search worst-case complexity?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

A junction/associative table.

2. Model answer

A key made from two or more attributes together.

3. Model answer

On a match or after examining all items.

4. Model answer

O(log n) for sorted random-access data.

By the end, you can…

• Replace 'Maths, Art' in one cell with separate enrolment records.

• Check whether key is composite, such as StudentID+CourseID.

• Move StudentName to Student and CourseTitle to Course.

Vocabulary — know these terms

1NF — Attributes hold atomic values and rows can be uniquely identified.

2NF — In 1NF, no non-key attribute depends on only part of a composite candidate key.

partial dependency — Non-key field depends on a subset of a composite key.

anomaly — Insertion, update or deletion causes undesirable effects.

NEW LEARNING

Read, annotate and explain

Make fields atomic

Replace 'Maths, Art' in one cell with separate enrolment records.

Identify keys

Check whether key is composite, such as StudentID+CourseID.

Remove partial dependencies

Move StudentName to Student and CourseTitle to Course.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Enrolment(StudentID,CourseID,StudentName,CourseName,Grade) has composite PK (StudentID,CourseID). Achieve 2NF.

StudentID+CourseID key

Find the given data and name the required result.

Names partly depend on key

Explain why this intermediate stage is necessary.

Split Student and Course

StudentName depends only on StudentID; CourseName only on CourseID. Split into Student(StudentID PK,StudentName), Course(CourseID PK,CourseName), Enrolment(StudentID FK,CourseID FK,Grade) with composite PK.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Explain atomicity in 1NF and correct a cell containing 'Red, Blue, Green'. [6]

Question 2

Explain partial dependency using OrderID+ProductID key and ProductDescription. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Convert StudentID,ClubID,StudentName,ClubName,JoinDate with composite key to 2NF. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A relation with a single-attribute candidate key can have a partial dependency on that key.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.